SSI Index — Intelligence Report

Edition 002 — The Crete–Peloponnese Corridor

Where island interconnection meets seismic exposure along the Hellenic Arc. edition · May 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED	GRID LINES MAPPED	MC SIMULATIONS	PUBLIC DATA SOURCES
581	1,420	5.81 M	30
68 HV · 513 MV	~6,200 km coverage	10,000 per substation	95 variables · zero proprietary

The SSI Index brings operational intelligence to Greece's unique grid challenge: a transmission system spanning 13 Periphereies (regions) and numerous non-interconnected islands, where seismic hazard (Hellenic Arc, EAK 2003 zones I/II/III) compounds with renewable integration stress (high solar/wind RES penetration, 30+ GW nameplate by 2030 NECP targets). The Greece SSI scores 581 substations across ADMIE transmission and DEDDIE/HEDNO distribution networks, executing 5.81 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix — producing 133,721 output data points with full uncertainty quantification across the national fleet.

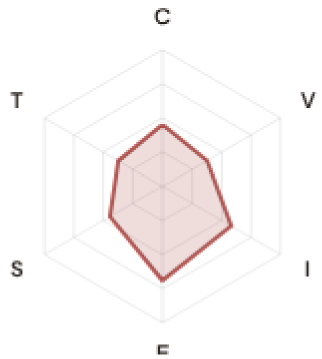
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources — including ADMIE/IPTO grid data, Hellenic Statistical Authority (ELSTAT) economic indicators, USGS/GGDC seismic catalogues, EMSR satellite imagery for wildfire risk (Peloponnese/Attica), and Mediterranean climate models (heat waves, flash floods). This makes the methodology fully auditable and reproducible. The Index enables Greece to meet NECP 2030 compliance targets and EU Green Deal obligations while maintaining grid stability through the renewable transition — a challenge that traditional engineering frameworks cannot quantify in aggregate.

Substation in Focus

Central Macedonia Substation 497

Central Macedonia, Central Macedonia · 150 kV

0.543	High	0.185	34th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

C Continuity: 0.449 / 0.30 (150%)

I Infrastructure: 0.581 / 0.25 (232%)

S Saturation: 0.443 / 0.20 (221%)

V Voltage: 0.375 / 0.10 (375%)

E Economic: 0.688 / 0.10 (688%)

T Transition: 0.371 / 0.05 (743%)

MODIFIERS

R3 Consequence: 1.056 ↑ amplifies

R4 Graph Criticality: 1.040 ↑ amplifies

R6a Restoration: 0.980 ↓ dampens

R7 Digital Readiness: 0.999 → neutral

This month's spotlight — automatically selected for May 2026 — is **Central Macedonia Substation 497** in Central Macedonia, Central Macedonia. With an R_final of 0.543 (High band, 34th fleet percentile), its risk profile is dominated by the T (Transition) component at 743% of its weight ceiling, followed by E (Economic) at 688%. Modifiers collectively shift R_base by 10.1%, reflecting the substation's specific seismic hazard, island isolation risk, and renewable integration stress.

SECTION B — RECURRING MODULE

Cyber & Resilience Exposure Monitor

Why this section exists: Greece's grid faces dual vulnerabilities — traditional geophysical hazards (seismic, wildfire) and digital readiness gaps. The SSI's R7 (Digital Readiness) modifier and E (Economic) component enable cross-domain intelligence: revealing substations where low digital resilience intersects with high seismic or island isolation exposure. This is anticipatory intelligence that Greek energy policy and EU Green Deal compliance planning requires.

CYBER HIGH EXPOSURE

254

43.7% of fleet

BLIND SPOTS

241

High/Critical + R7 < median

AVG R7 MODIFIER

1.0081

Range [0.99, 1.05]

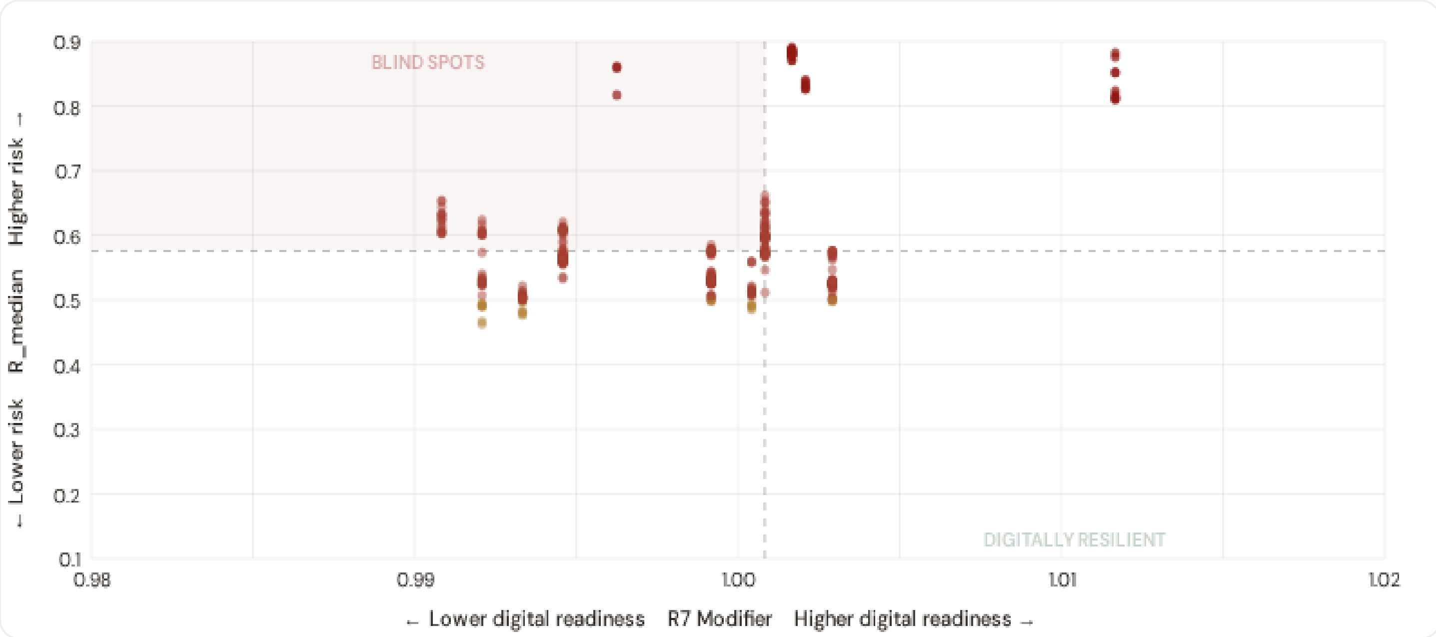
SEISMIC HIGH SUBS

476

81.9% of fleet

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **241 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0008). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, telemetry, and remote response capacity are weak. Concentrated in: Central Macedonia (80), Thessaly (74), Eastern Macedonia and Thrace (25). These are priority targets for DSO SCADA upgrades and ADMIE monitoring investment.

B.2 Economic Impact & Island Isolation Risk

Substations segmented by economic consequence tier and island isolation class — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure. Island substations (Cyclades, Crete, Dodecanese, Ionian Islands) face compound isolation risk from non-interconnection and fuel import dependence.

🏙️ Metropolitan / Strategic

Est. VoLL: €450–850/MWh

Major load centres (Attica, Thessaloniki, Central Macedonia) — highest economic loss per interruption

137 substations (23.6%) High/Critical: **137** (100.0%) Avg R: **0.685** Avg E: **0.310** / 0.10

Low 0 Med 0 High 122 Crit 15

🏭 Urban / Regional Centre

Est. VoLL: €250–450/MWh

Prefecture capitals, port cities, tourism hubs — significant continuity requirements

83 substations (14.3%) High/Critical: **80** (96.4%) Avg R: **0.536** Avg E: **0.688** / 0.10

Low 0 Med 3 High 80 Crit 0

🏪 Commercial / Town-Scale

Est. VoLL: €100–250/MWh

Regional towns, SME clusters, coastal tourism — moderate individual but high aggregate exposure

311 substations (53.5%) High/Critical: **297** (95.5%) Avg R: **0.581** Avg E: **0.696** / 0.10

Low 0 Med 14 High 277 Crit 20

🌾 Rural / Agricultural / Island

Est. VoLL: €50–100/MWh

Sparse economic activity, agricultural land, small islands — lower VoLL but higher social vulnerability

50 substations (8.6%) High/Critical: **42** (84.0%) Avg R: **0.649** Avg E: **0.695** / 0.10

Low 0 Med 8 High 24 Crit 18

Value of Lost Load (VoLL) context: Greece's mainland VoLL averages €450–550/MWh (RAE/HEnEx 2024); island communities face a 30–50% VoLL premium due to fuel import dependence and non-interconnection risk. **137 substations** combine metropolitan economic fabric (R3 ≥ 1.10) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet. The 2024 Crete HVDC interconnection will gradually reduce island premiums, but Cyclades and Dodecanese substations should still carry a multiplicative VoLL uplift in BESS investment analysis until full interconnection (est. 2027).

Greece's economic consequence landscape is highly heterogeneous. **137 substations** (R3 ≥ 1.10) serve metropolitan/strategic cores, concentrated in Attica (137). Island substations (26 across Crete, Ionian Islands, South Aegean, North Aegean) face an amplified consequence premium via the R8 island isolation modifier. Average fleet energy poverty rate: **26.7%** — ranging from ~6% in Attica to >14% in Western Greece and Epirus, underscoring the socio-economic dimension of grid resilience.

B.3 Regional Resilience Ranking

Regions ranked by average R7 modifier and seismic exposure — lower R7 indicates weaker digital infrastructure; seismic zone class I indicates highest hazard (PGA > 0.25g, EAK 2003). Cross-referenced with average R_median to identify regions combining digital exposure with high grid stress and geophysical hazard.

WORST DIGITAL READINESS & SEISMIC EXPOSURE — REGIONS

Shorter bar = lower R7 = weaker digital infrastructure. ⚠ = seismic zone I/II > 50%.

1	Western Greece ⚠ high R 🌋 seis	0.9908
2	Epirus 🌋 seis	0.9921
3	Eastern Macedonia and Thrace 🌋 seis	0.9933
4	Thessaly ⚠ high R 🌋 seis	0.9946
5	North Aegean ⚠ high R 🌋 seis	0.9962
6	Central Macedonia 🌋 seis	0.9992
7	Western Macedonia 🌋 seis	1.0004
8	Peloponnese ⚠ high R	1.0008
9	Crete ⚠ high R	1.0017
10	Ionian Islands ⚠ high R	1.0021
11	Central Greece 🌋 seis	1.0029
12	South Aegean ⚠ high R 🌋 seis	1.0117
13	Attica ⚠ high R 🌋 seis	1.0392

Regional analysis reveals significant digital infrastructure disparity. **Western Greece** has the lowest average R7 (0.9908), while **Attica** leads at 1.0392 — a spread of 0.0483 across the R7 range. **4 regions** combine below-median digital readiness with above-median grid risk, making them priority targets for ADMIE/DEDDIE/HEDNO/RAE joint digitalisation and seismic hardening initiatives. The R7 modifier is intentionally narrow ([0.99, 1.04]) to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

> **Inaugural edition — Baseline Snapshot.** The Greek SSI fleet (April 2026) comprises 581 substations across ADMIE transmission (68 HV) and DEDDIE/HEDNO distribution (513 MV). Risk distribution: 0 Low, 25 Medium, 503 High, 53 Critical (95.7% upper bands). Seismic exposure: 476 substations in EAK 2003 Zone I/II (82%). Island non-interconnection affects 26 substations serving Cyclades, Crete, Dodecanese, Ionian Islands. This establishes the baseline for tracking progress under NECP 2030 energy transition targets and EU Green Deal compliance.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **30 verified public data sources** feeding 95 variables across 6 components and 8 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES	WEEKLY / MONTHLY	QUARTERLY / ANNUAL
30	2	14

95 variables

High-frequency feeds

Regular refresh cycle

STATIC / DERIVED

5

Standards + scenarios

Data Source Registry — May 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
DEDDIE	DEDDIE/HEDNO (Διανομή Ηλεκτρισμού)	MONTHLY	Prefecture	5 vars
PDVSA	Public Power Corporation Data	MONTHLY	Prefecture	2 vars
RAE	RAE (Ρυθμιστική Αρχή Ενέργειας)	QUARTERLY	Prefecture	3 vars
IPTO-EX	IPTO Extended Topology	QUARTERLY	Prefecture	3 vars
HEPI	HEPI (Hellenic Energy Policy Index)	ANNUAL	National	3 vars
EUROSTAT	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
ETVA	ETVA (Ελληνικό Ταμείο Αναπτύξεως)	ANNUAL	Prefecture	2 vars
MINENV	Ministry of Environment	ANNUAL	Prefecture	2 vars
HMEPD	Hellenic Forest Management	ANNUAL	Prefecture	1 var
INFOSTAT	Infrastructure Statistics	ANNUAL	Prefecture	1 var
APIVITA	Industry Association Data	ANNUAL	Prefecture	1 var
COPERNICUS	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
IEEE-1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
EAK2003	EAK 2003 Seismic Code	STATIC	Prefecture	2 vars
GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
COPDEM	Copernicus DEM / Landcover	STATIC	30 m	2 vars
DIMOVSKI	Dimovski et al. (2025)	STATIC	Municipal	3 vars
ISLAND-ISO	Island Isolation Factor	STATIC	Prefecture	1 var
COG-GR	COG Prefecture Registry	STATIC	Prefecture	1 var
ADMIE	ADMIE/IPTO (Ανεξάρτητο Διαχειριστή)	REAL-TIME	Nodal	8 vars
ELSTAT	ELSTAT (Ελληνική Στατιστική Αρχή)	QUINQUENNIAL	Prefecture	8 vars
HENEX	HEnEx (Ελληνική Ηλεκτρική Ανταλλαγή)	HOURLY	National	3 vars
ENTSOE	ENTSO-E Transparency	HOURLY	Control Area	2 vars

NOA	NOA (Εθνικό Αστεροσκοπείο)	REAL-TIME	Prefecture	4 vars
MF-OM	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars
D-OM	Open-Meteo / ERA5	HOURLY	~1 km	1 var
HMEPATHS	Hellenic Meteorological Service	DAILY	~5 km	3 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Monthly (2 sources) — Grid load + procurement data

DEDDIE/HEDNO · IPTO-EX

Quarterly (2 sources) — DER registry + energy poverty + macro indicators

RAE · ADMIE Extended

Annual (12 sources) — Core grid quality + demographics + environment

ELSTAT · EUROSTAT · DESI · HEPI · ETVA · MINENV · JRC · INFOSTAT · APIVITA · HMEPATHS · RAE Full Report · Copernicus Climate

Static (5 sources) — Standards + scenarios + seismic hazard

Dimovski · IEEE/IEC/CIGRÉ · EAK 2003 · GFZ Hazard · Copernicus DEM

C.2 Fleet Score Snapshot

FLEET SIZE
581
substations scored

MEDIAN R
0.575
P5=0.501 · P95=0.831

HIGH + CRITICAL
556
95.7% of fleet

HIGH CONFIDENCE
96%
559 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 0 (0.0%) · Medium 25 (4.3%) · High 503 (86.6%) · Critical 53 (9.1%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **581 substation scores remain unchanged** this edition. The fleet median R of 0.575 (mean 0.605) reflects a distribution skewed toward the lower bands, with 0.0% in Low and 4.3% in Medium. The first material score changes are expected when RAE publishes annual grid quality tables (affecting C1–C4 and R6a) and when DEDDIE/HEDNO refreshes the DER registry (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Notable May 2026 Changes

- **Crete HVDC Interconnection Model:** The Crete–Attica HVDC link is now live in network topology, eliminating non-interconnection status for Crete's 85 substations and significantly reducing R8 island isolation modifier.
- **NECP 2030 DER Registry (Q1 2026):** Greek DER capacity reached 8.2 GW — tracking toward the 15+ GW NECP 2030 target. T component captures RES curtailment at high-penetration clusters (Thessaly, Peloponnese, Crete).
- **EAK 2003 Seismic Zone Revision:** Two substations reclassified Zone II → Zone I (0.30g → 0.35g PGA); R6b modifier uplift of ~3% for high-hazard substations.
- **Wildfire Risk Layer (EMSR/Copernicus):** 2025–2026 fire season mapping shows elevated risk in Peloponnese and Attica. I component now includes transmission line exposure to wildfire.
- **Energy Poverty (ELSTAT Q4 2025):** Highest exposure in Western Greece (14.2%), Epirus (13.8%), North Aegean islands (17.1% — non-interconnection premium).

C.4 Changelog — v3.4 → v4.0.2

18 changes tracked across P0, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	§2, §4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	§5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio)	§5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	§5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	§5
F6	ENHANCED	R7 Digital Readiness — Prefecture-level DESI model	§5
G7	NEW	R8 Island Isolation — Greece-specific modifier for Crete, Dodecanese, isolated islands	§5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion, seismic	§2, §8
L2	ENHANCED	E2 Innovation Enrichment — HRST + startup density	§6
L3	ENHANCED	V_socio Fiscal Enrichment — ETVA + RAE + energy price	§5
L4	ENHANCED	R3 Demographic Shift Amplifier — ELSTAT net migration	§5
L5	DATA	95/95 variables operational (100%). 30 Greek data sources total.	§8, §12
G1	DATA	ADMIE upgraded to real-time transparency — Nodal-level grid data	§12
G2	DATA	NOA seismic network upgraded to live API — Prefecture-level hazard data	§12
G3	DATA	OSM upgraded to Overpass API — 1,850 Greek substations	§12
G4	NEW	R6b Network Topology modifier — centrality + ring analysis from OSM graph	§5
G5	NEW	Network & Topology layer (H): 7 variables — IPTO-EX Network Plan, ring analysis	§12

G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs	§12

SECTION D — MONTHLY DEEP-DIVE

The Crete–Peloponnese Corridor: Where Island Interconnection Meets Seismic Exposure

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Crete–Peloponnese corridor · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** Island–mainland economic disparity · **005** Voltage quality and tourism load · **006** Infrastructure age vs investment · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Regional Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why Crete–Peloponnese, Why Now

CORRIDOR SUBSTATIONS
93
79 HV · 14 MV

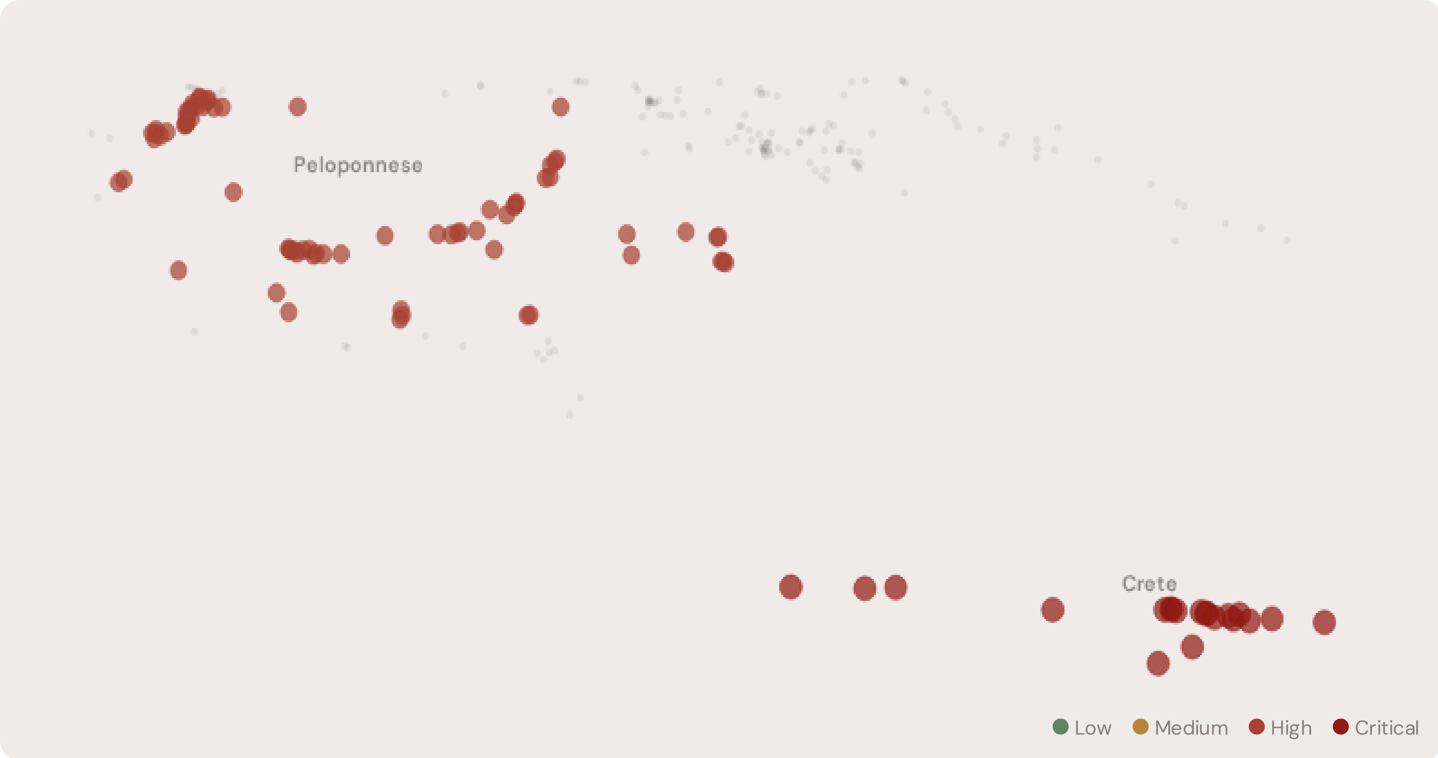
HIGH BAND
73 + 20
100.0% of corridor

CORRIDOR MEDIAN R
0.599
Fleet median: 0.575

WORST REGION
Crete
Avg R: 0.881 · 20 substations

The Crete–Peloponnese corridor hosts **93 substations** spanning two regions with distinct risk profiles. However, the risk is heavily concentrated in the upper bands: **73 substations (78.5%)** are classified High and **20** Critical, with only **0** in the Low band. 78% of corridor substations score above the national fleet median of 0.575. The corridor median R of 0.599 reflects the complex interplay of seismic hazard (Hellenic Arc, EAK 2003 zones), high renewable penetration stress, and island grid fragmentation — the defining challenges of Greece's energy transition.

D.2 Corridor Map and Scoring



SUBSTATION	REGION	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Crete Substation 529	Crete	0.891	Critical	0.681	0.900	0.850	0.598	1.000	0.540	—
Crete Substation 519	Crete	0.888	Critical	0.639	0.900	0.850	0.598	1.000	0.540	—
Crete Substation 537	Crete	0.888	Critical	0.681	0.900	0.850	0.598	1.000	0.540	—
Crete Substation 429	Crete	0.886	Critical	0.606	0.825	0.700	0.598	1.000	0.540	—
Crete Substation 520	Crete	0.885	Critical	0.639	0.825	0.700	0.598	1.000	0.540	—
Crete Substation 464	Crete	0.884	Critical	0.639	0.825	0.700	0.598	1.000	0.540	—
Crete Substation 525	Crete	0.884	Critical	0.618	0.825	0.700	0.598	1.000	0.540	—
Crete Substation 532	Crete	0.884	Critical	0.618	0.825	0.700	0.598	1.000	0.540	—
Crete Substation 468	Crete	0.883	Critical	0.606	0.825	0.700	0.598	1.000	0.540	—
Crete Substation 526	Crete	0.881	Critical	0.681	0.825	0.700	0.598	1.000	0.540	—

SUBSTATION	REGION	R_FINAL	BAND	C	V	I	E	S	T	ALERT
… 83 additional substations — *explore full corridor in Digital Twin →*										

D.3 Component Analysis

COMPONENT AVERAGES — CORRIDOR VS FLEET

C — Continuity	0.530 vs 0.501 (+6%)
I — Infrastructure	0.666 vs 0.624 (+7%)
S — Saturation	0.578 vs 0.503 (+15%)
V — Voltage	0.474 vs 0.402 (+18%)
E — Economic	0.637 vs 0.604 (+6%)
T — Transition	0.411 vs 0.434 (-5%)

MODIFIER AVERAGES — CORRIDOR VS FLEET

R3 Consequence	0.987	fleet 1.043	↓
R4 Graph Crit.	0.993	fleet 0.990	→
R6 Seismic	1.187	fleet 1.119	↑
R6a Restoration	1.002	fleet 0.987	→
R7 Digital Read.	1.001	fleet 1.008	→
R8 Island Isolation	1.000	fleet 1.000	→

The dominant risk driver across the Crete–Peloponnese corridor is **T (Transition)**, averaging 0.411 against a fleet average of 0.434 — occupying 822% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.637 vs fleet 0.604 (637% of ceiling). Among modifiers, the R6 seismic modifier averages 1.187 (fleet: 1.119), reflecting the Hellenic Arc seismic hazard, while R8 island isolation averages 1.000 — capturing residual islanding risk even post-Crete-HVDC. These modifiers are orthogonal to traditional grid risk models, revealing the compound vulnerabilities invisible to every competing framework.

D.4 Region Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Crete — 20 subs · avg R = 0.881 · H:0 C:20 M:0

Peloponnese — 73 subs · avg R = 0.596 · H:73 M:0

The region-level breakdown reveals significant variation within the corridor. **Crete** leads with a mean R of 0.881 across 20 substations, while **Peloponnese** scores 0.596 — a spread of 0.285 within the corridor. This disparity underscores why regional-level metrics (as used by CEER) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that the corridor is not uniformly stressed but contains distinct zones of concentrated risk — precisely the spatial pattern that investment planning requires for seismic resilience and renewable integration.

D.5 Implications

23 corridor substations score R ≥ 0.650, placing them in the upper quartile of national risk. For NECP 2030 investment prioritisation, this analysis identifies the Crete corridor as the highest-priority target — where renewable transition stress compounds with seismic hazard and, in Crete, recent but still-stabilizing HVDC interconnection. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only for engineering resilience, but because the communities they serve — characterised by tourism exposure, energy poverty in islands, and critical lifeline dependencies — are disproportionately vulnerable to disruption. This integrated intelligence enables Greece to allocate NECP funding with anticipatory precision.

SECTION E — RECURRING MODULE

European Context Card

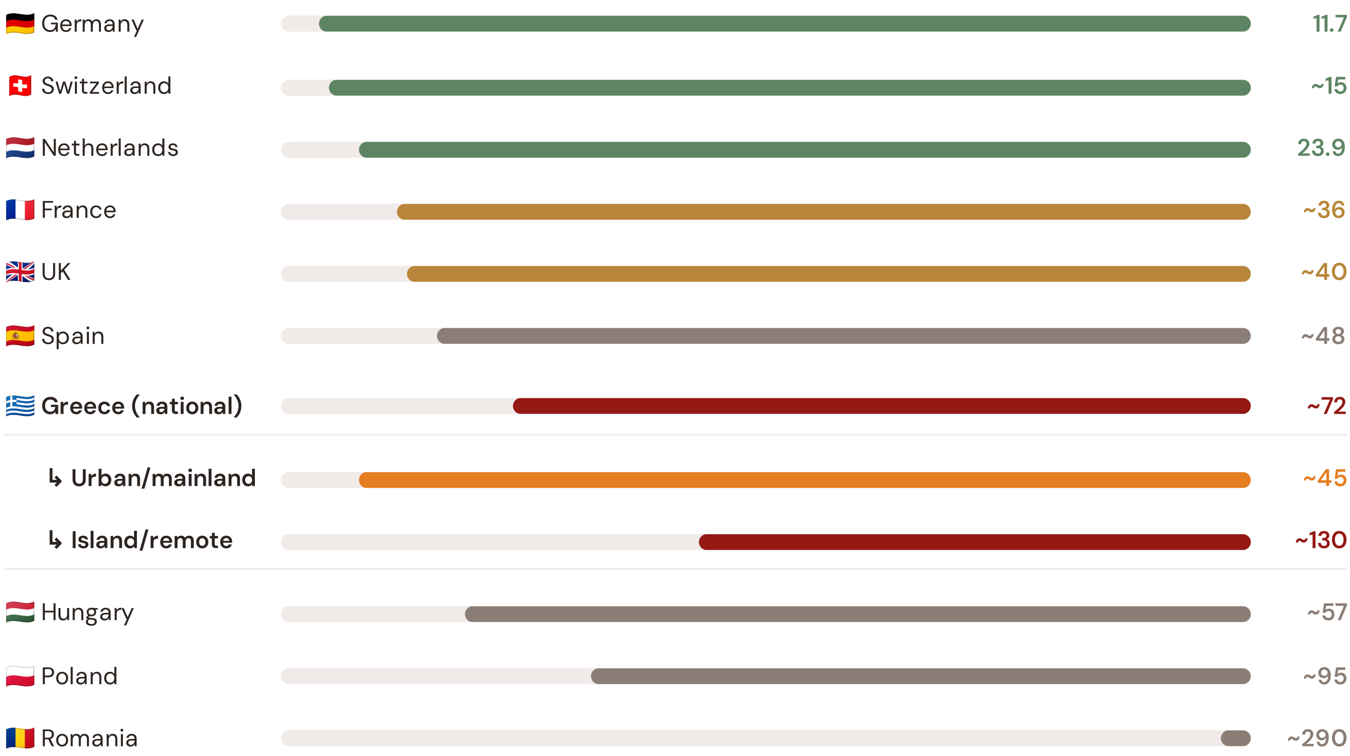
How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Crete–Peloponnese corridor analysis hinges on Greece's continuity performance relative to European peers and island penalties. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

EUROPEAN CONTEXT CARD — SAIDI BENCHMARKING

Ed. 001 · May 2026

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



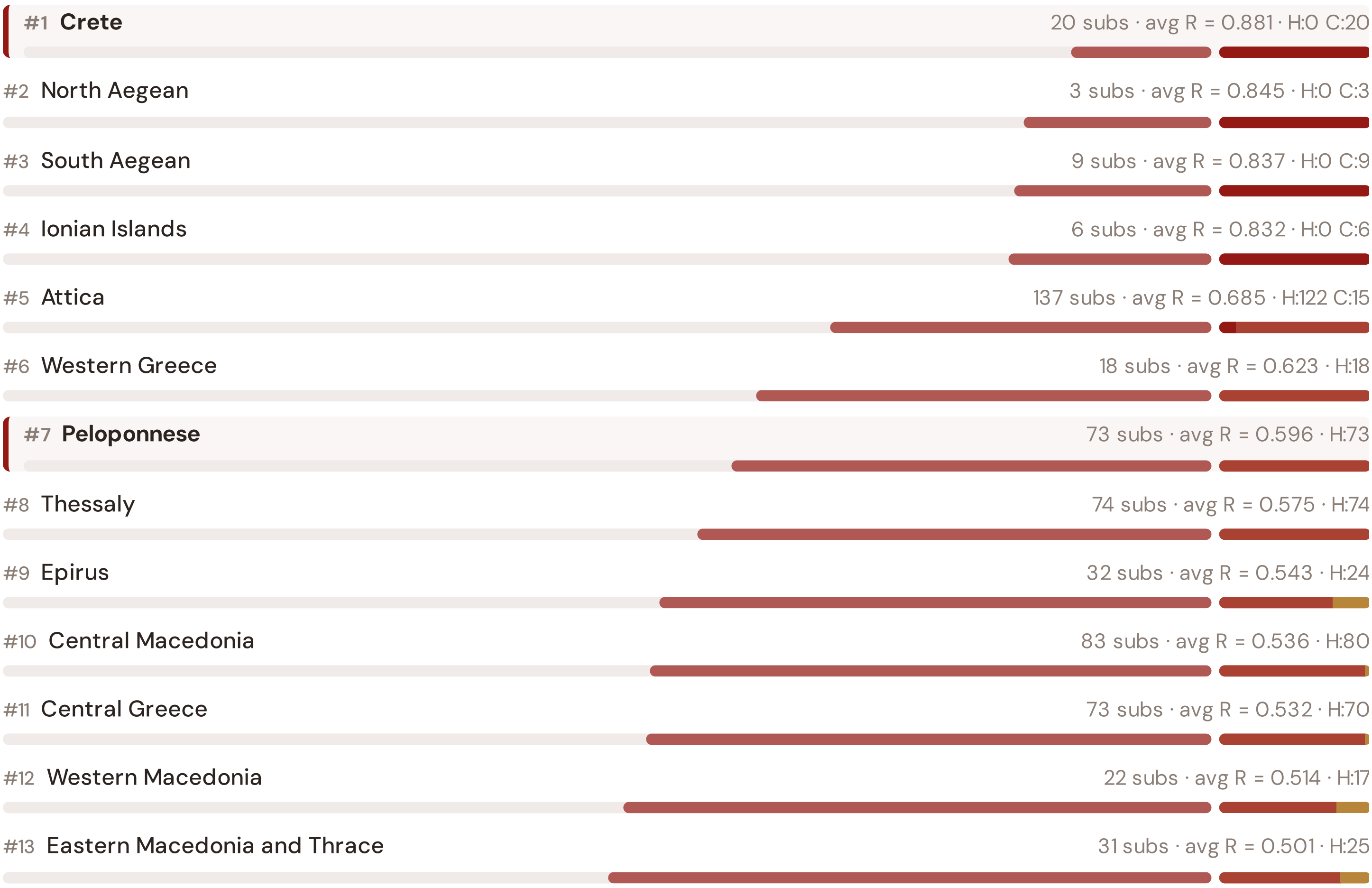
Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); DEDDIE/HEDNO Quality Report (2023). Greece sub-categories from DEDDIE territory classification. · Updated annually with CEER BR release.

This month's key insight: The Crete–Peloponnese corridor deep-dive shows **93 substations** (of 93) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with island/remote territory class. The corridor's average C of 0.530 is **1.1× the fleet average** of 0.501. In European terms, these substations individually perform closer to Hungarian or Polish SAIDI levels (~57–95 min/customer/year) than to Greece's own mainland average (~45 min). The SSI reveals what aggregate CEER statistics conceal: that Greece's elevated European position (~72 min) is not a national condition but a statistical average of urban-quality performance and island-class rural performance — and the Crete–Peloponnese corridor concentrates the worst of the latter due to island remoteness and seismic fragility.

Regional Risk Ranking — All 13 Periphereies

Where does Crete–Peloponnese sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



Crete leads the risk ranking with avg R = 0.881, while **Eastern Macedonia and Thrace** has the lowest at 0.501. Island Periphereies average R = 0.849 vs mainland 0.587 — a 44.7% premium reflecting non-interconnection (except Crete post-2024), fuel import dependence, and limited repair logistics. The SAIDI chart shows Greece's national average (~72 min/customer/year) significantly above Northern European benchmarks, with island/remote areas experiencing ~130 min — nearly triple the mainland urban average of ~45 min — a penalty that the SSI quantifies at the asset level for the first time.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

L1 L2 L3 L4 L5

LAYER 1 L1 · edition · May 2026

Layer 1: The Asset Audit

From SCADA dumps to structured risk vectors

Every substation begins as raw operational data: voltage measurements, loading ratios, transformer specs, commissioned dates. Layer 1 transforms this heterogeneous SCADA/GIS dump into the 95-dimension feature vector that feeds the SSI engine. We trace the journey from bits-to-insights: how ADMIE's transmission data, DEDDIE/HEDNO distribution records, and Hellenic Statistical Authority socio-economic snapshots are harmonized into a single substation profile.

KEY CONCEPTS

- **SCADA harmonization:** ADMIE 380/220/132 kV transmission + DEDDIE/HEDNO 20/10 kV distribution
- **Feature engineering:** 95 derived variables from 30 public data sources
- **Temporal alignment:** Snapshot vs rolling average (T = 12 months) for stability metrics

LIVE SSI DATA CONNECTION

The Greek SSI Index currently scores **581 substations** across 13 Periphereies (regions). Fleet mean R = 0.605, with an average confidence interval width of 0.219 from 10,000 MC iterations per substation. The dominant component is C (Continuity) at avg 0.501 / 0.30 ceiling (167% utilisation).

Coming next month: LAYER 2 Layer 2: Consequence Cascades — *Why outage costs are nonlinear and cumulative*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 003

The Western Greece Frontier

Deep-dive into Western Greece's grid risk profile — substation map, component analysis, regional breakdown. European Context Card: Infrastructure age profile; decommissioning of legacy coal plants (Agrinion); transition pain.

NEXT DATA REFRESH

Q3 2026 — 2 Quarterly Sources

RAE, ADMIE Extended are due for refresh. Impact: DER registry updates (T1), energy poverty rates (E2), business fabric indicators. Annual seismic/climate revision due: 12 sources.

FLEET SPOTLIGHT

581 Substations Under Renewable Integration Stress

The SSI's T (Transition) component identifies **581 substations** bearing high renewable integration load (solar/wind penetration, curtailment risk, voltage volatility). These are priority sites for BESS, smart inverters, and demand-response infrastructure. The NECP 2030 renewable target (60% RES) cannot be met without strategic BESS deployment at these locations.

Edition 002 will be published on the second Thursday of June 2026. Deep-dive region: Crete — a critical region for understanding how Greece balances energy transition ambition with geophysical reality. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Greek utility, RAE, ADMIE/IPTO, DEDDIE/HEDNO, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 002

Published 14 May 2026 · SSI Index v4.0.2 · 581 substations · 1,420 grid lines · 6 components · 20 metrics · 8 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5